



CWI

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*Centrum Wiskunde & Informatica
Eindhoven University of Technology*

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Are we modeling the wrong stress in LES?

In collaboration with Benjamin Sanderse and Roel Verstappen



Stress in LES

S. D. Agdestein

Turbulence

Lifecycle of turbulence

Distribution of energy

LES components

Discrete LES

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The Burgers' equation (1D)

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$$\partial_t u = \underbrace{\nu \partial_{xx} u}_{\text{Diffusion}} - \underbrace{\partial_x (u^2/2)}_{\text{Convection}}$$



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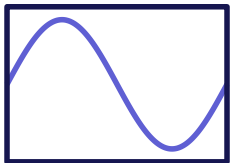
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Effect on a Fourier mode

$$u = e^{ikx}$$





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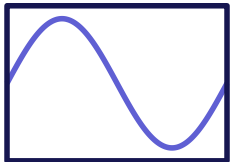
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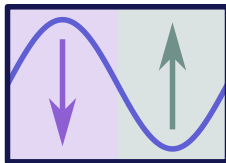
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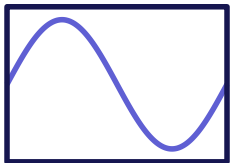
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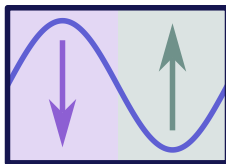
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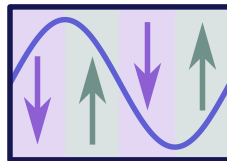
$$u = e^{ikx}$$



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$$\partial_x (u^2/2) = ike^{i(2k)x}$$





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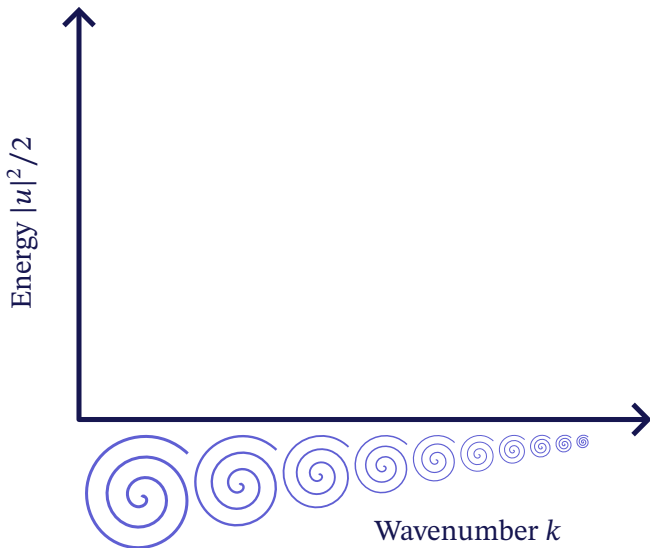
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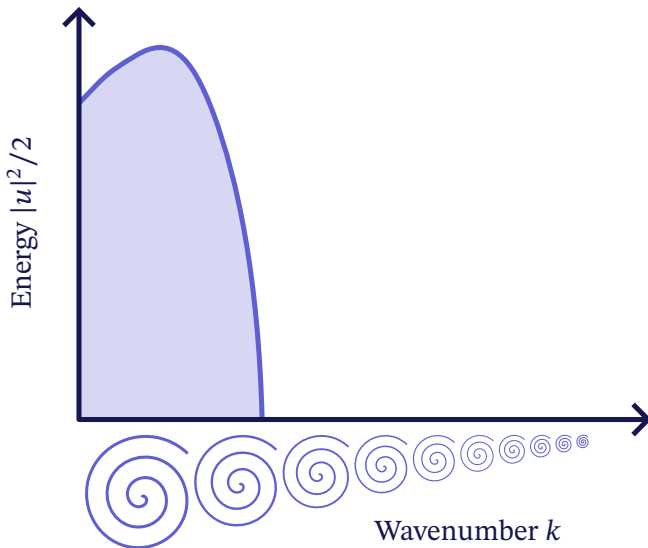
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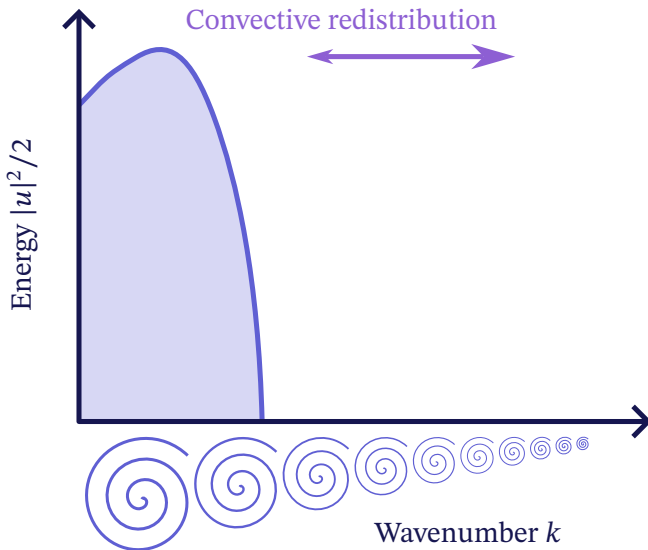
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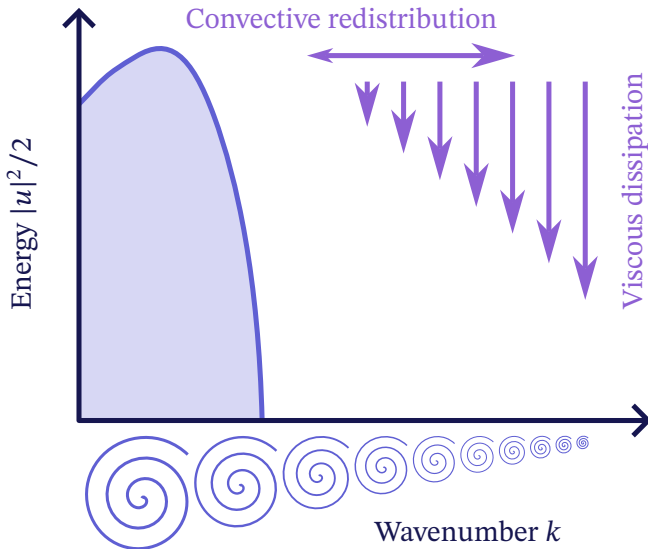
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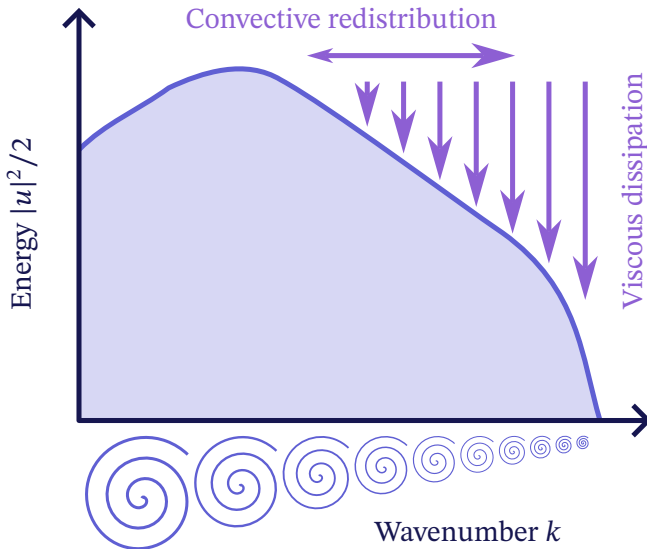
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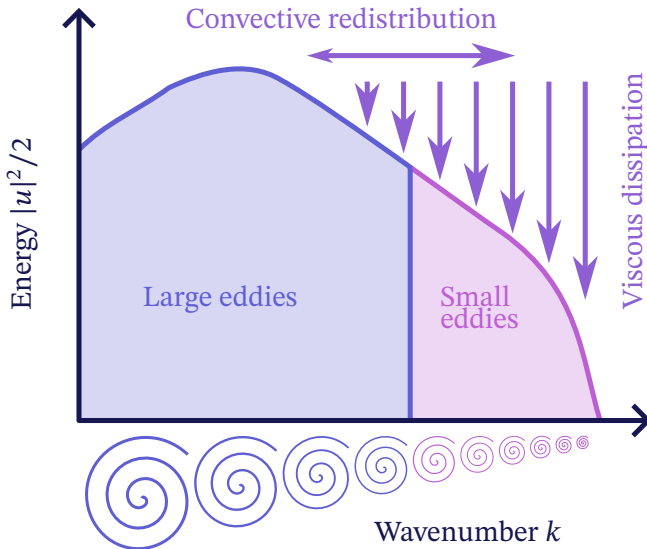
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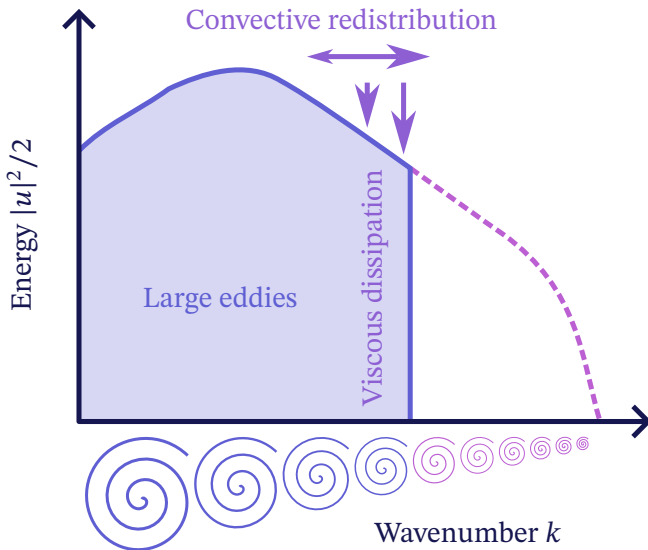
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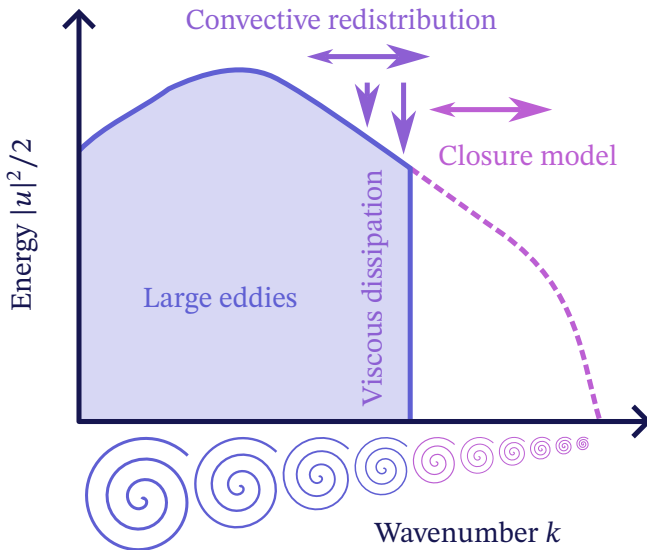
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LES and FVM: the toolbox

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Filtering

$$\overline{(\cdot)}^{\Delta}, \overline{(\cdot)}^h$$

LES and FVM: the toolbox



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Divergence theorem

$$\partial_x^h = \overline{(\cdot)}^h \circ \partial_x$$



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$$\tau^{\Delta}(u) \approx m^{\Delta}(\bar{u}^{\Delta})$$

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Pressure projection

$$\operatorname{div} \circ \pi = 0, \quad \pi \pi = \pi$$

Divergence theorem

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Time integration

$$u^* = u^n + \Delta t \dots$$
$$u^{n+1} = \pi u^*$$



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LES filter

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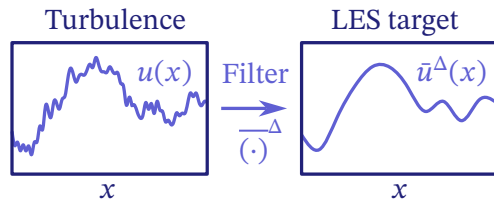
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$$\bar{u}^{\Delta}(x) = \int_{\Omega} G^{\Delta}(x - y)u(y) \, dy$$



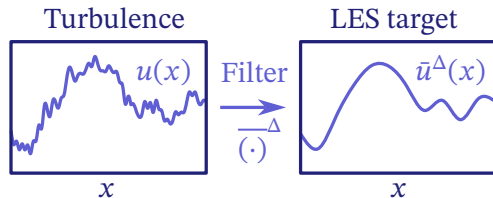
$$\bar{u}^\Delta(x) = \int_{\Omega} G^\Delta(x - y) u(y) dy$$





LES filter

$$\bar{u}^{\Delta}(x) = \int_{\Omega} G^{\Delta}(x-y)u(y) \, dy$$



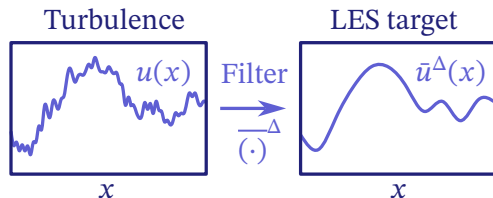
FVM filter

$$\bar{u}^h(x) = \frac{1}{h} \int_{x-h/2}^{x+h/2} u(y) \, dy$$



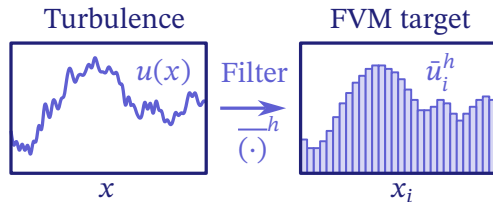
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The mechanism powering the FVM



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The mechanism powering the FVM

$$\partial_x^h u(x) = \frac{u\left(x + \frac{h}{2}\right) - u\left(x - \frac{h}{2}\right)}{h} = \frac{1}{h} \int_{x-h/2}^{x+h/2} \partial_y u(y) dy = \overline{\partial_x u}^h(x)$$



Divergence theorem

The mechanism powering the FVM

$$\partial_x^h u(x) = \frac{u\left(x + \frac{h}{2}\right) - u\left(x - \frac{h}{2}\right)}{h} = \frac{1}{h} \int_{x-h/2}^{x+h/2} \partial_y u(y) dy = \overline{\partial_x u}^h(x)$$

The discrete divergence ∂_x^h is induced by the FVM filter $\overline{(\cdot)}^h$
 $\Rightarrow$ **exact** discrete conservation laws for $\bar{u}^h$



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Closure modeling

Models rely on τ^Δ being the unresolved stress



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Models rely on τ^Δ being the unresolved stress

$$\tau^\Delta(u) = \overline{uu}^\Delta - \bar{u}^\Delta \bar{u}^\Delta, \quad S(u) = (\nabla u + \nabla u^T)/2$$



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Structural models

$$m^\Delta(\bar{u}^\Delta) = \frac{\Delta^2}{12} (\nabla \bar{u}^\Delta)(\nabla \bar{u}^\Delta)^T, \quad (\text{Taylor series truncation})$$

$$m^\Delta(\bar{u}^\Delta) = \tau^{(2\Delta)}(\bar{u}^\Delta), \quad (\text{scale-similarity})$$



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Functional models

$$m^\Delta(\bar{u}^\Delta) = -2\nu^\Delta S(\bar{u}^\Delta), \quad 2\nu_{\text{ideal}}^\Delta = -\frac{S(\bar{u}^\Delta) : \tau^\Delta(u)}{S(\bar{u}^\Delta) : S(\bar{u}^\Delta)}, \quad (\text{eddy-viscosity})$$



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Infinitesimal flux

Flux reconstruction

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$$\sigma(u) = uu$$



Infinitesimal flux

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$$\sigma(u) = uu$$

Discrete flux

$$\sigma^h(u)(x) = \left[\frac{u\left(x + \frac{h}{2}\right) + u\left(x - \frac{h}{2}\right)}{2} \right]^2 \quad (\text{energy-conserving central})$$
$$\sigma^h(u)(x) = \begin{cases} u\left(x - \frac{h}{2}\right)^2, & \text{if } u\left(x - \frac{h}{2}\right) > 0, \\ u\left(x + \frac{h}{2}\right)^2, & \text{if } u\left(x + \frac{h}{2}\right) < 0, \\ 0, & \text{otherwise,} \end{cases} \quad (\text{dissipative upwind})$$

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Projection and time integration

Incompressible Navier-Stokes as a conservation law



Projection and time integration

Incompressible Navier-Stokes as a conservation law

With pressure and constraint

$$\nabla \cdot u = 0, \quad \partial_t u + \nabla \cdot (\sigma(u) + p\delta) = f, \quad \sigma(u) = uu - \nu(\nabla u + \nabla u^T)$$

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Pressure-free formulation

$$\partial_t u + \nabla \cdot \pi \sigma(u) = \pi f$$



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Pressure-free formulation

$$\partial_t u + \nabla \cdot \pi \sigma(u) = \pi f$$

Time integration

$$\begin{aligned} u^* &= u^n + \Delta t(f - \nabla \cdot \sigma(u^n)) \\ u^{n+1} &= \pi u^* \end{aligned}$$



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Correcting for an inconsistency



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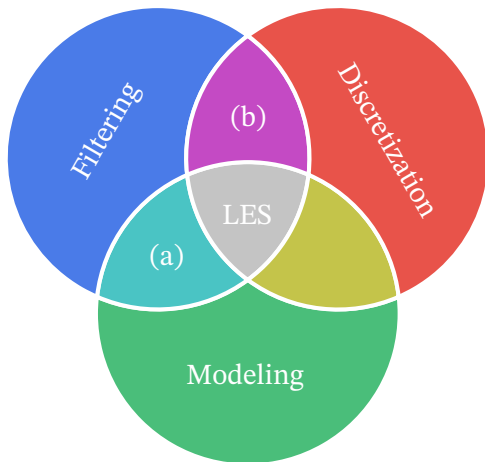
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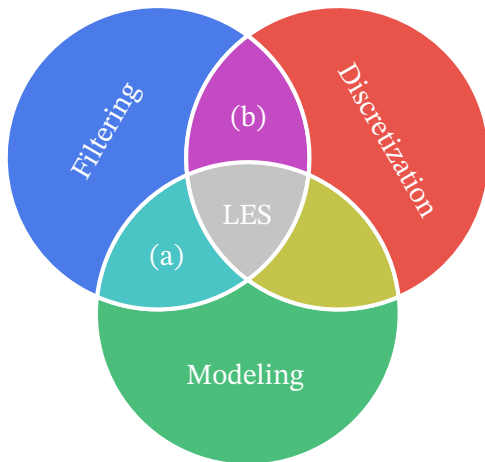
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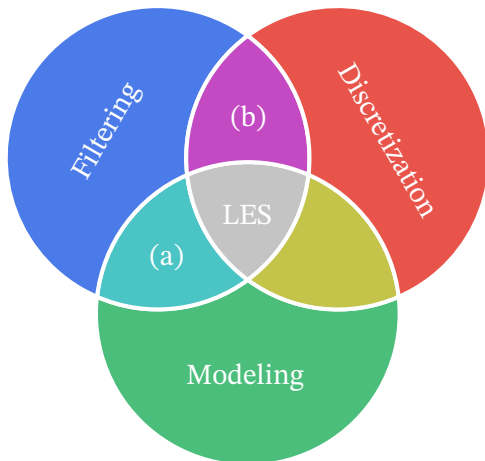
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Discretization-aware LES

Correcting for an inconsistency



- a** Classical approach:
Filter and model,
then discretize.



Discretization-aware LES

Correcting for an inconsistency

- a** Classical approach:
Filter and model,
then discretize.
- b** New approach:
Filter and discretize,
then model.



Burgers' equation

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$$\partial_t u + \partial_x \sigma(u) = 0, \quad \sigma(u) = uu/2 - \nu \partial_x u$$



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Apply LES/FVM filters

$$\partial_t u + \partial_x \sigma(u) = 0, \quad \sigma(u) = uu/2 - \nu \partial_x u$$

$$\partial_t \bar{u}^{\Delta, h} + \partial_x^h \overline{\sigma(u)}^{\Delta} = 0$$

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Apply LES/FVM filters

$$\partial_t u + \partial_x \sigma(u) = 0, \quad \sigma(u) = uu/2 - \nu \partial_x u$$

$$\partial_t \bar{u}^{\Delta,h} + \partial_x^h \overline{\sigma(u)}^\Delta = 0$$

Introduce numerical flux

$$\partial_t \bar{u}^{\Delta,h} + \partial_x^h \sigma^h(\bar{u}^{\Delta,h}) = -\partial_x^h \tau^{\Delta,h}(u)$$



$$\partial_t u + \partial_x \sigma(u) = 0, \quad \sigma(u) = uu/2 - \nu \partial_x u$$

Apply LES/FVM filters

$$\partial_t \bar{u}^{\Delta,h} + \partial_x^h \overline{\sigma(u)}^\Delta = 0$$

Introduce numerical flux

$$\partial_t \bar{u}^{\Delta,h} + \partial_x^h \sigma^h(\bar{u}^{\Delta,h}) = -\partial_x^h \tau^{\Delta,h}(u)$$

Unresolved discrete flux

$$\tau^{\Delta,h}(u) = \overline{\sigma(u)}^\Delta - \sigma^h(\bar{u}^{\Delta,h})$$



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Incompressible Navier-Stokes



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Projection form

$$\partial_t u = -\nabla \cdot \pi \sigma(u), \quad \sigma(u) = uu - \nu (\nabla u + \nabla u^T)$$

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Projection form

$$\partial_t u = -\nabla \cdot \pi \sigma(u), \quad \sigma(u) = uu - \nu (\nabla u + \nabla u^T)$$

Exact LES equations

$$\text{Infinitesimal LES: } \partial_t \bar{u} = -\overline{\nabla \cdot \pi \sigma(u)} = -\nabla \cdot \overline{\pi \sigma(u)}$$

$$\text{Discrete LES: } \partial_t \bar{u}^h = -\overline{\nabla \cdot \pi \sigma(u)}^h = -\nabla_h \cdot \overline{\pi \sigma(u)}^{h,*}$$

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τ_{11}	τ_{12}	τ_{13}
τ_{12}	τ_{22}	τ_{23}
τ_{13}	τ_{23}	τ_{33}

τ_{11}	τ_{12}	τ_{13}
τ_{21}	τ_{22}	τ_{23}
τ_{31}	τ_{32}	τ_{33}



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Discretization contribution

Discretization-aware closure

DNS-aided LES

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Discretization contribution

How much of the sub-filter flux comes from the discretization?



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How much of the sub-filter flux comes from the discretization?

$$\tau^{\Delta,h}(u) = \tau_{\text{classic}}^{\Delta,h}(u) + \tau_{\text{flux}}^{\Delta,h}(u) + \tau_{\text{div}}^{\Delta,h}(u)$$



Discretization contribution

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Discretization contribution

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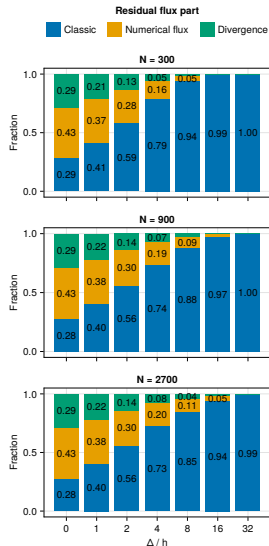
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Fitting the Smagorinsky constant for Burgers' equation

$$m(u, \theta) = -\theta s(u), \quad m^h(u, \theta) = -\theta s^h(u),$$

$$s(u) = |\partial_x u| \partial_x u, \quad s^h(u) = |\partial_x^h u| \partial_x^h u,$$



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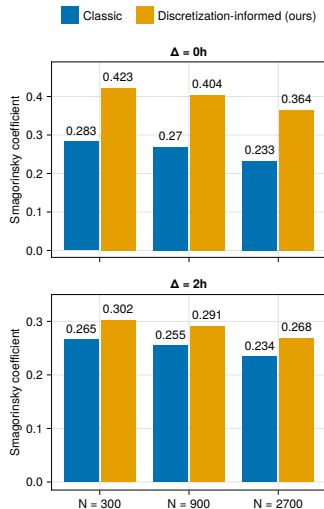
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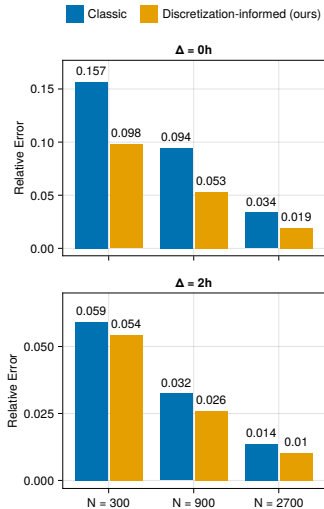
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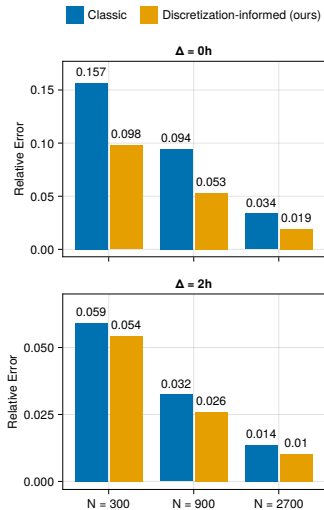
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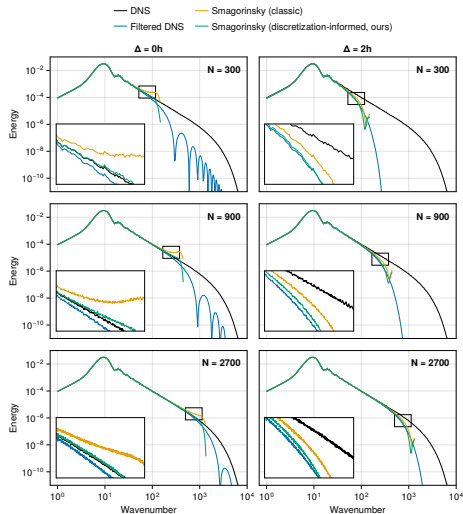
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Stress expressions for incompressible Navier-Stokes



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LES equation

$$\partial_t \bar{u}^h = -\nabla_h \cdot \pi^h [\sigma^h(\bar{u}^h) + \tau^h(u)]$$

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- τ_D^h : Correct SFS
- τ_E^h : Correct SFS (symmetrized)

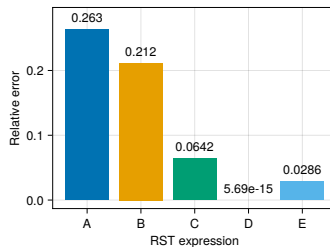


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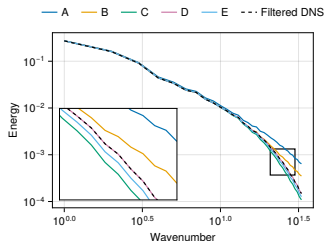
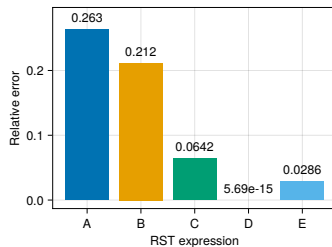


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Thank you!

The discretization is a fundamental part of LES.



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The discretization is a fundamental part of LES.

*For incompressible Navier-Stokes, the LES-FVM stress tensor is **non-local** and **non-symmetric**.*



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The discretization is a fundamental part of LES.

*For incompressible Navier-Stokes, the LES-FVM stress tensor is **non-local** and **non-symmetric**.*

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Implicit and explicit LES unified into one coherent framework.



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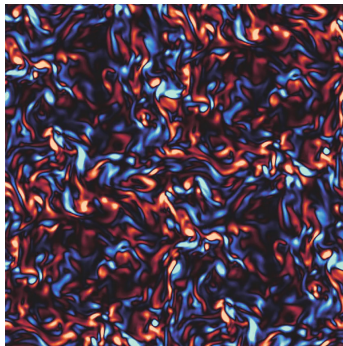
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<https://agdestein.github.io/>